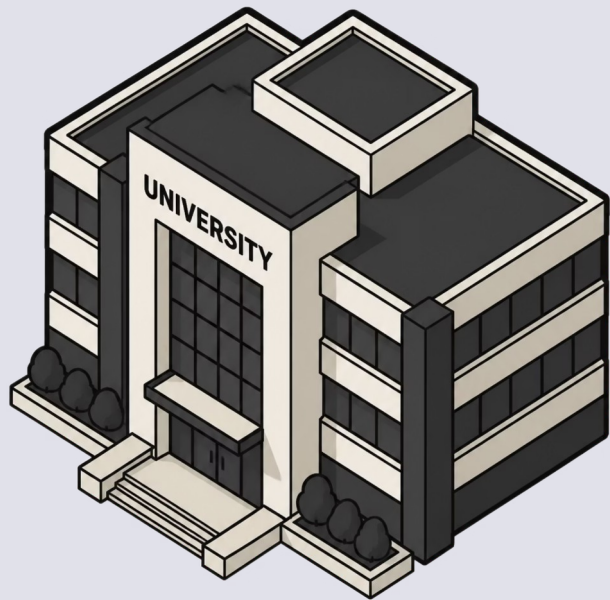
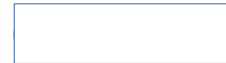




Campus Complaint Tracker

A Production-Ready Web Application for Campus Infrastructure & Service Request Management



The Problem



Lost Complaints

Manual processes with no paper trail



No Visibility

Zero transparency on resolution status



No Accountability

No analytics, no audit trail



The Solution



→ Centralized Platform

Role-based web application for
all stakeholders

→ Automated Reporting

Instant insights and data exports

→ Real-Time Tracking

Full transparency at every step



Core Features

Stateless JWT Auth

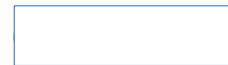
Secure, token-based authentication on every request

Dynamic JPA Filtering

Multi-criteria specification queries at runtime

Excel / CSV Reports

One-click report generation via Apache POI



User Roles

Students & Faculty

- Submit new complaints
- Track resolution status
- Edit or withdraw submissions

Administrators

- Assign complaints to staff
- Update and escalate status
- Export analytics reports



Tech Stack



Backend

Java 17 · Spring Boot 3.5 · MySQL 8



Security

Spring Security · JWT · BCrypt

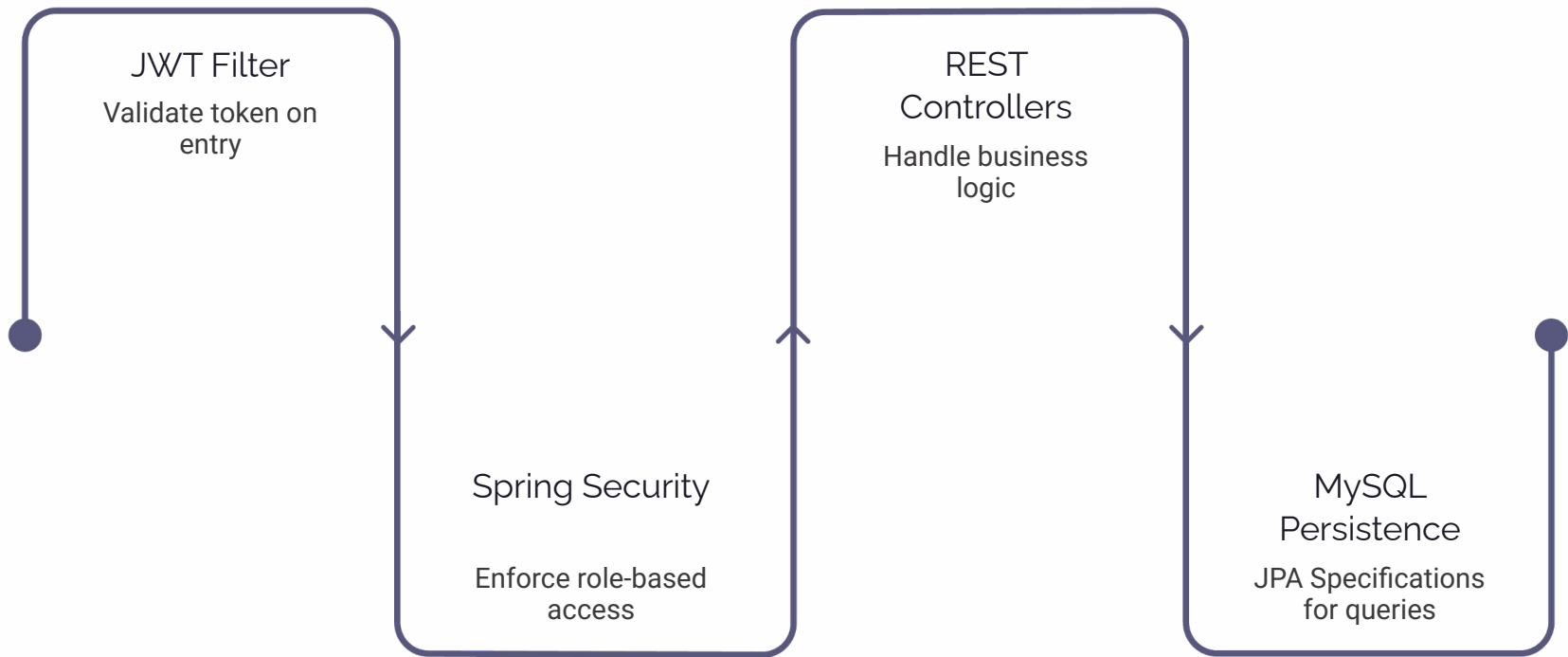


Tooling

Apache POI · Swagger UI



System Architecture



Every incoming request is authenticated, authorized, and processed through a clean layered architecture before touching the database.



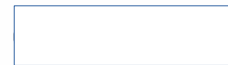
Database Design

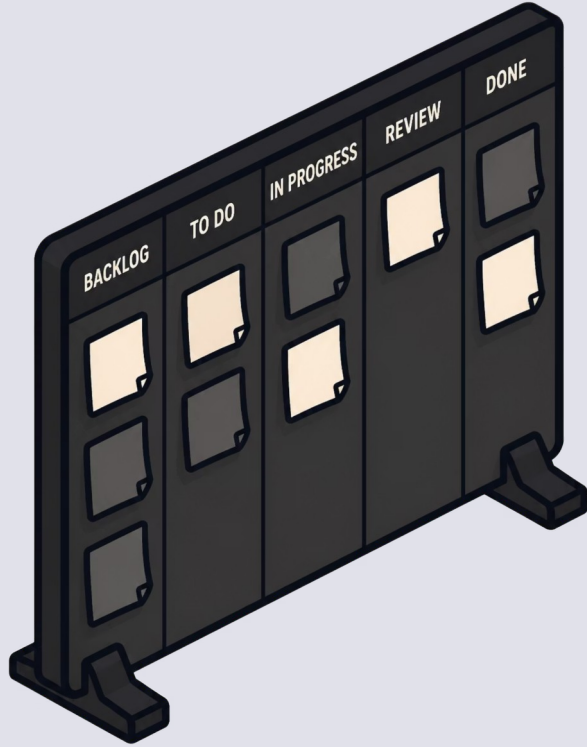
User Entity

Field	Type
id	PK / UUID
email	VARCHAR
password	BCrypt hash
role	ENUM
department	VARCHAR

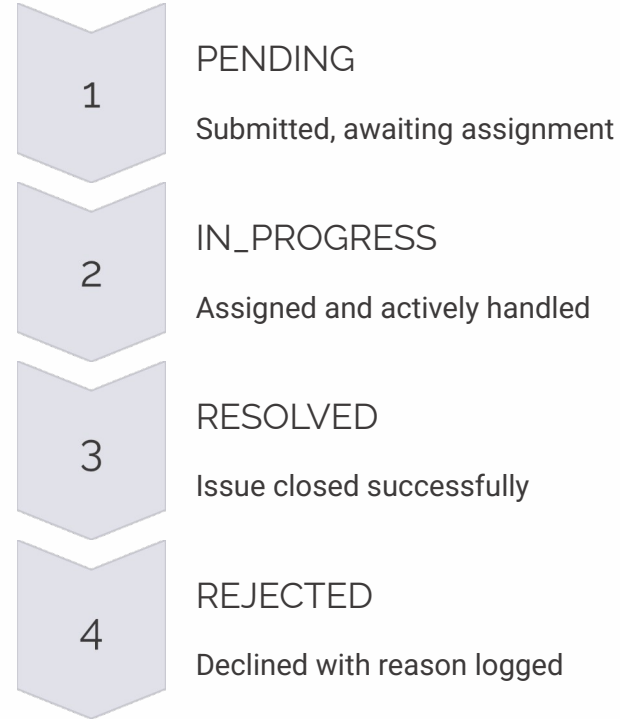
Complaint Entity

Field	Type
title	VARCHAR
category	ENUM
priority	ENUM
status	ENUM
assignedTo	FK → User





Complaint Lifecycle



Security & Dynamic Filtering

Security Layer

Stateless JWT Tokens

No server-side session state

BCrypt Hashing

Passwords never stored in plain text

Bearer Authorization

Role-based route protection on every endpoint

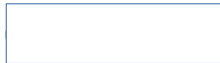
Dynamic Filtering

JPA Specifications

Multi-criteria queries
built at runtime

Filter Dimensions

Category · Priority ·
Location · Status · Date





Thank You

Campus Complaint Tracker — **Java 17** · **Spring Boot 3.5** · **JWT** · **JPA Specifications** · **MySQL 8** · **Apache POI**

 Secure

Stateless JWT + BCrypt

 Transparent

Real-time lifecycle tracking

 Scalable

Clean layered architecture

